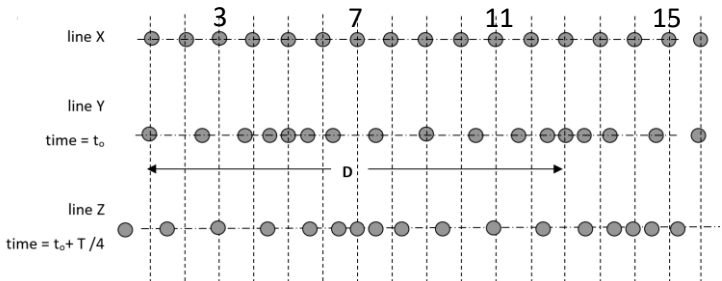
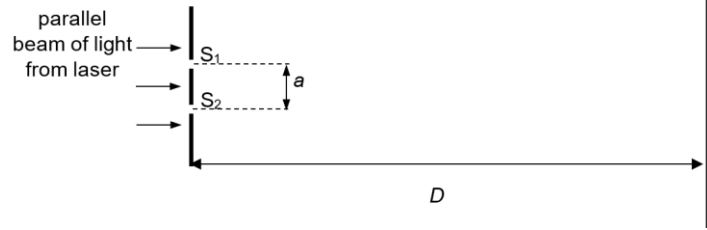


6	(a)	Particles of longitudinal waves vibrate along the direction of propagation of the wave. Particles of transverse waves vibrate perpendicular to the direction of propagation of the wave. 1 mark if: Longitudinal waves cannot be polarised by transverse waves can be.	B1
	(b)	(i) The <u>vibration is along the motion of the waves</u> . Hence cannot define the plane of polarisation	B1
		(ii) $D = 1 \frac{1}{2}$ wavelength	A1
		(iii) Graph Y: 1 mark for 2 cycles shown 1 mark for correct position of zero displacement -1 mark if negative sine graph (iv) Graph Z : 1 mark for Y shifted $\frac{1}{4} T$ – judge by starting point(ECF) 1 mark for filling in the first $\frac{1}{4}$ cycle Award only 1 mark if graph is shifted by $\frac{1}{4}$ cycle in the wrong direction	B1 B1 B1 B1

		(v)	 1 mark for compression shown corresponding to crossing of graph Z. 1 mark for rarefaction shown corresponding to crossing of graph Z. Award only 1 mark if compression and rarefaction are shown but sign convention is wrong.	B1 B1
	(c)	(i)	 Diagram showing correct a and D. (a must be centre to centre or equivalent)	A2
		(ii)	$a = \lambda D/x$ $= (620 \times 10^{-9})(2)/(0.002)$ $= 6.2 \times 10^{-4} \text{ m}$ $= 6.2 \times 10^{-1} \text{ mm}$	M1 A1 A1
	(d)	(i)	Intensity decreases (of the peak) and Separation of dark fringes (width of the pattern) increases.	B1 B1
		(ii)	$\theta = \tan^{-1} (0.13/2.8) = 2.658^\circ$ $\lambda = b \sin \theta = 13 \times 10^{-6} \sin 2.658^\circ$ $= 6.03 \times 10^{-7} \text{ m}$	M1 M1 A1